Exercise 03: SQL Foundations
SQL CASE Statements

1) The products Table

SQL Query 01

```
SELECT product_name,
       price,
       Case
           WHEN price > 1000 THEN 'Expensive'
           WHEN price BETWEEN 100 AND 1000 THEN
                'Mid_range'
           ELSE 'Budget'
       END AS price_category
FROM products;
```

Output:

product_name	price	price_category
Laptop	1200.00	Expensive
Phone	800.00	Mid-range
Keyboard	45.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

2) The orders Table

SQL Query 2

```
SELECT customer_name,
       amount,
       CASE
           WHEN amount >= 1000 THEN 'High Value'
           WHEN amount BETWEEN 500 AND 999.99
                THEN 'Medium Value'
```

```
    ELSE 'Low Value'
  END AS order_value_category
FROM orders;
```

Output:

Customer_name	amount	order_value_category
Alice	150.00	Low Value
Bob	560.00	Medium Value
Charlie	999.99	Medium Value
Diana	45.50	Low Value
Ethan	1200.00	High Value

3) The employees Table

SQL Query 03

```
SELECT emp_name,
       department,
       salary,
       CASE
         WHEN department = 'IT' AND salary > 85000 THEN
              'SENIOR IT'
         WHEN department = 'HR' AND salary > 55000
              THEN 'Experienced HR'
         ELSE 'Staff'
       END AS position_level
FROM employees;
```

Output:

emp_name	department	Salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

4) The students Table

SQL Query 4

```
SELECT student_name,
       score,
       CASE
           WHEN score >= 90 THEN 'A'
           WHEN score BETWEEN 80 AND 89 THEN 'B'
           WHEN score BETWEEN 70 AND 79 THEN 'C'
           WHEN score BETWEEN 60 AND 69 THEN 'D'
           ELSE 'F'
       END AS grade
FROM students;
```

Output:

student_name	Score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

5) The deliveries Table

SQL Query 05

```
SELECT delivery_id,
       delivery_time_minutes,
       CASE
           WHEN delivery_time_minutes <= 30 THEN
                'Fast'
           WHEN delivery_time_minutes BETWEEN
                31 AND 60 THEN 'On Time'
           ELSE 'LATE'
       END As performance
FROM deliveries;
```

Output:

delivery_id	delivery_time_minutes	performance
1	45	On time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

6) The tickets Table

SQL Query 6

```
SELECT issue_type,
       priority,
       CASE priority
           WHEN 3 THEN 'High'
           WHEN 2 THEN 'Medium'
           WHEN 1 THEN 'Low'
       END AS priority_label
FROM tickets;
```

Output:

issue_type	priority	priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

7) The attendance Table

SQL Query 7

```
SELECT student_id,
       (day_present * 100 / total_days) AS
           attendance_percentage,
       Case
```

```
        WHEN (days_present * 100/total_days) >=
            90 THEN 'Excellent'
        WHEN (days_present * 100/total_days) BETWEEN
            75 AND 89 THEN 'Good'
        ELSE 'Needs Improvement'
    END As attendance_status
FROM attendance;
```

Output:

student_id	attendance_percentage	attendance_status
1	90	Excellent
2	60	Needs Improvement
3	96	Excellent
4	50	Needs Improvement
5	100	Excellent

9) The products_inventory Table

SQL Query 8

```
SELECT product_id,
    stock_qty,
    CASE
        WHEN stock_qty = 0 THEN 'Out of Stock'
        WHEN stock_qty BETWEEN 1 AND 5
            THEN 'Low Stock'
        ELSE 'In Stock'
    END AS stock_status
FROM products_inventory;
```

Output:

product_id	stock_qty	stock_status
1	5	Low Stock
2	0	Out of Stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

9) The classes Table

SQL Query 9

```
SELECT subject,
       enrolled_students,
       CASE
           WHEN enrolled_students >= 25 THEN 'Large'
           WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
           ELSE 'Small'
       END AS class_size_category
FROM classes;
```

Output:

subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

10) The payments Table

SQL Query 10

```
SELECT payment_id,
       payment_method,
       amount,
       CASE
           WHEN payment_method = 'cash' AND amount >= 200 THEN 'Eligible for Discount'
           ELSE 'Not Eligible'
       END AS discount_eligibility
FROM payments;
```

Output:

payment_id	payment_method	amount	discount_eligibility
1	Card	50.00	Not Eligible
2	Cash	200.00	Eligible for Discount
3	Card	150.00	Not Eligible
4	PayPal	75.00	Not Eligible
5	Cash	300.00	Eligible for Discount